Implicit Differentiation
(Section 3.7)

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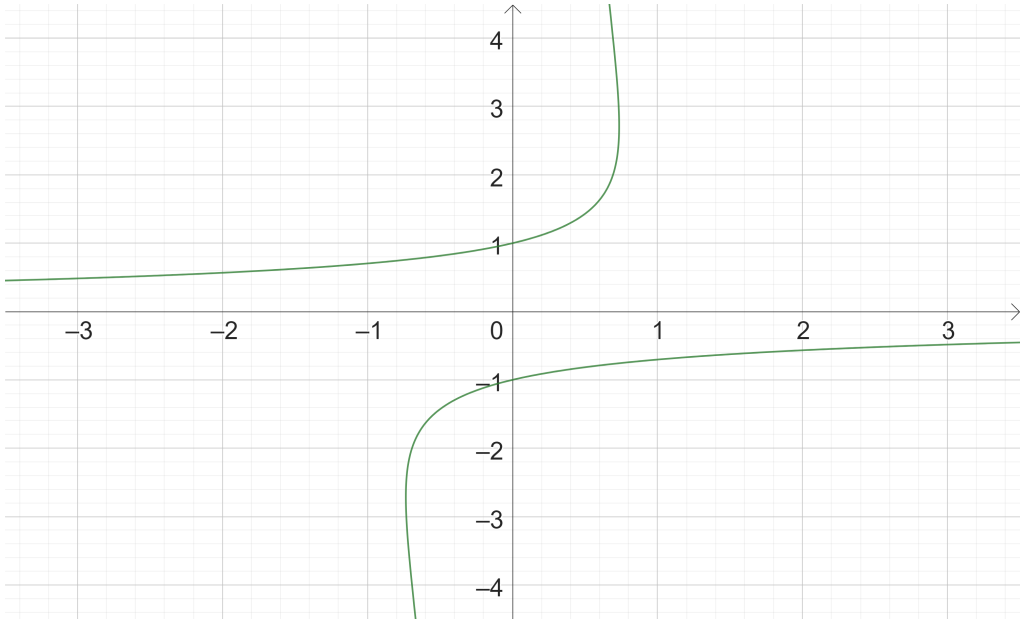
MATH 2413

When Do We Use It?

Remark

We've found the ways to compute most derivatives of functions. But what if y is not related to x with a (unique) function? If we know that $y^2 = e^{xy}$, can we find $\frac{dy}{dx}$ without solving for y?



Steps

Remark

The steps for implicit differentiation are as follows:

- *Take the derivative of both sides of your equation with respect to x (or whatever independent variable you're dealing with).*
- *Treat y as a differentiable function of x, and use chain rule as necessary.*
  - *For example, $\frac{d}{dx}\sin(y) = \cos(y) \cdot \frac{dy}{dx}$.*
- *Solve for $\frac{dy}{dx}$. The answer will likely involve both x and y.*

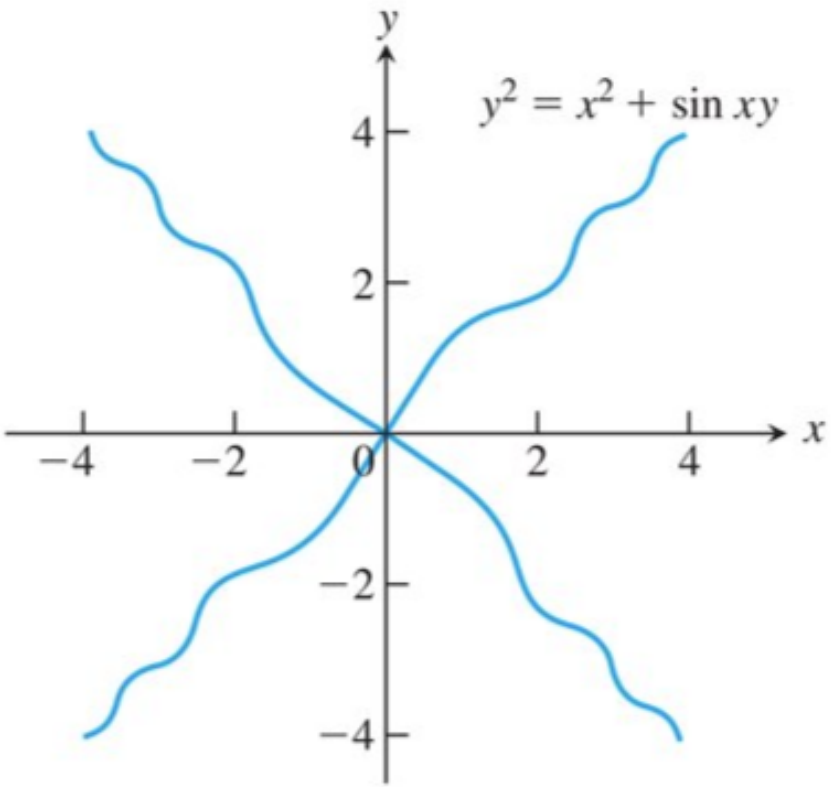
Example

Let $y^2 = e^{xy}$. Find $\frac{dy}{dx}$.

Another Example

Example

Let $\pi^2 y^2 = x^2 + \sin(xy)$. Find $\frac{dy}{dx}$.



Derivatives at Points

Remark

Sometimes we'll be interested in $\frac{dy}{dx}$ at particular (x, y) points. If both x and y are given, then simply plug them into $\frac{dy}{dx}$. If only one is given, you may need to go back to the original equation.

Example

Let $\pi^2 y^2 = x^2 + \sin(xy)$. Find $\frac{dy}{dx}$ at $(x, y) = (\pi, 1)$

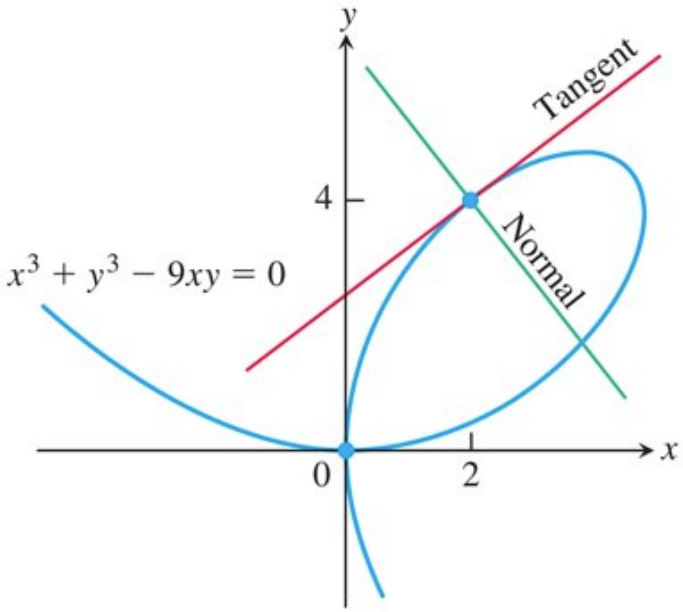
Example

Let $y^2 = e^{xy}$. Find $\frac{dy}{dx}$ when $y = 1$.

Tangent Lines

Remark

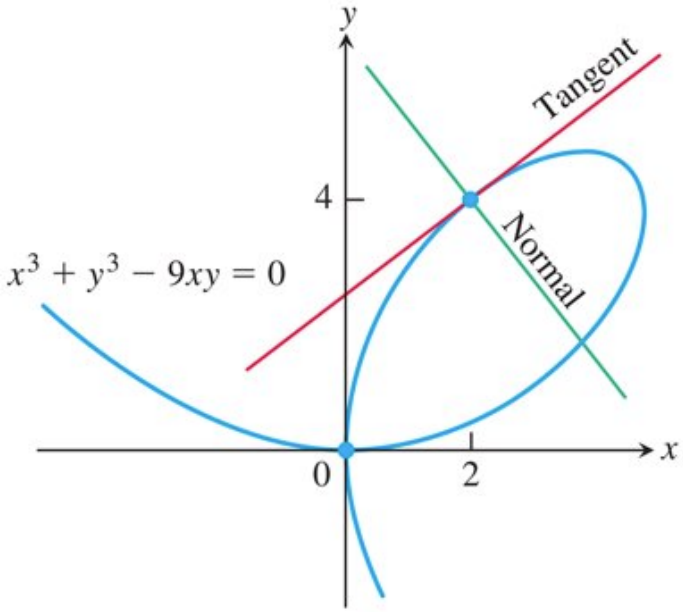
With this method, finding tangent lines on graphs like these is now possible. In fact, the process is effectively the same as before! Note: A ***normal*** *line is simply the line perpendicular to the tangent line at the tangent point, having a slope of* $\frac{-1}{dy/dx}$.



Example

Example

Let $x^3 + y^3 - 9xy = 0$. Find the equations of the tangent and normal lines at the point (2,4).



More Examples

Example

Find the equations of the tangent and normal lines for the following curves at the indicated points.

- $ye^{x^2} = (y + x)(y + 2x)$, $P = (0, 1)$
- $x = e^y$, $P = (e, 1)$
- $x = \sin(y)$, $P = (1/2, \pi/6)$